LONGEVITY 06

MOTS-c

16-amino-acid peptide coded in mitochondrial DNA; products are made synthetically



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Peptide	Insulin resistance, obesity, aging	Lab or animal only	Not FDA approved · compounding allowed

STUDIED FOR

Insulin resistance	Body fat	Exercise capacity
ANIMAL	ANIMAL	ANIMAL

Bone	Artery calcification
ANIMAL	ANIMAL

No human studies. FDA found no published study giving MOTS-c to people. One Phase 2a trial in prediabetes is registered, with no results yet.

EVIDENCE & WHAT'S KNOWN

EVIDENCE

LAB OR ANIMAL
LEVEL 1 OF 5

- Mouse, rat and cell studies since 2015
- No published human study, any route
- No animal toxicity studies found

NOT YET ESTABLISHED

- Safety in people
- Half-life in people
- Dose in people
- Immune reactions to injections

HOW IT WORKS IN THE BODY

MOTS-c ▶ Folate cycle ▶ AMPK switched on ▶ Muscle metabolism

- **Slowed the folate cycle** and purine production, in cells
- **Switched on AMPK,** an energy-sensing enzyme; muscle the main target (lab)
- **Broken down quickly** in human blood in a test tube
- **Its molecular target** is unknown (FDA)

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established in people	None studied in people

PHARMACOKINETICS	SOLD AS
Not established in people	Compounded (Rx) or "research use only"

MONITORED IN TRIALS	SPORT (WADA)
Not established (no human studies)	Prohibited, class S4

REGULATORY

FDA	Not approved

HHS	Compounding allowed, acetate, SC or IM (Oct 2026)

WADA	Prohibited at all times (S4.4.1)

REFERENCES

1. FDA, Pharmacy Compounding Advisory Committee briefing on MOTS-c, July 2026
2. HHS, Federal Register notice 2026-22192, Oct 30, 2026
3. WADA, 2026 Prohibited List

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

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